Step 1 — Grid generation per capita, Nigeria 2010–2025

Status: complete. Feeds correction plan Phase 1.6, Step 1. **Workbook:** `step1_per_capita_diagnostic.xlsx`

Question

Does observed growth in Nigerian grid generation carry information about growth in electricity **demand**, or only about **population growth** and the persistence of supply constraints?

This matters because Phase 1.6 must source a demand-growth rate `g`, and the obvious candidate — historical generation CAGR — is only admissible if the series reflects demand rather than delivery.

Result

Generation growth decomposes multiplicatively as `(1 + g_gen) = (1 + g_pop) × (1 + g_percap)`.

Window	Generation	Population	Per capita	Demographic share
2010–2025	3.14%	2.39%	**0.73%**	76%
2013–2025	3.06%	2.29%	**0.76%**	75%
2015–2025	2.28%	2.22%	**0.05%**	98%
2017–2025	3.22%	2.16%	**1.04%**	67%
2014–2024	1.53%	2.27%	**−0.73%**	>100%

Per-capita grid generation moved from ~157 kWh/person (2010) to ~175 kWh/person (2025). Eighteen kWh per person over fifteen years is **2.1 W of additional continuous supply per person** — and under two of the five windows it is flat or negative.

Population growth is stable at 2.2–2.4% across every window. Per-capita growth is not stable: it ranges from −0.73% to +1.04% depending on endpoints, which is the signature of a series driven by supply events rather than by a demand trend.

NERC-basis figures (2022–2025 only) give a higher per-capita rate (1.2–1.8%/yr), but rest on two to three annual observations and are reported for comparison only.

Interpretation

Between 67% and more than 100% of observed generation growth is demographic. The residual is small, unstable, and window-dependent.

The series therefore carries almost no signal about consumption intensity. Using it as an organic demand-growth rate would assume the supply constraint persists, and then report modest capacity requirements *because* demand was assumed not to grow — which is circular. This is the same error class as the superseded

23.08 TWh demand base: a quantity produced by the system's dysfunction being read as a statement about underlying demand.

The historical series does have a legitimate role: it defines a `constrained_continuation` arm — "what if the constraints persist" — which is the pessimistic bound, not the central case.

External corroboration. NIRP 2024 (Table 14) derives its national demand forecast bottom-up from population, GDP, electrification rate and loss trajectories, explicitly rather than from historical generation. The national planning authority reached the same methodological conclusion independently.

Values carried into `scenarios.py`

Arm	Rate	Basis
`constrained_continuation` central	**3.14%**	Ember 2010–2025 generation CAGR
range floor	1.53%	weakest window (2014–2024)
range ceiling	3.22%	strongest window (2017–2025)

Label these as a **constraint-persistence** arm. They are not organic demand growth and must never be presented as such.

Sources

Series	Source	Role
Grid generation, level	NERC quarterly reports, gross generation sent out at busbar	**Authoritative.** Sets the 37.09 TWh (2024) Tier 1 anchor
Grid generation, long series	Ember Electricity Data Explorer, Nigeria annual generation	**Shape only.** Not used for level
Population	World Bank WDI `SP.POP.TOTL` (UN WPP + national statistical office)	Denominator

Flags — read before using any number here

[F1] Population figures are projections. Nigeria's last census was 2006. The projections are contested, and the uncertainty propagates directly into every per-capita figure.

[F2] NERC 2018 Q4 and 2019 Q1 are transcribed as the identical value (8,878,473 MWh). One is a transcription error. Neither year is used until this is resolved against the source reports.

[F3] The NERC 2022 entry is an average-power figure (3,988 MWh/h), not an annual energy total. Converted as × 8,760 h. Whether it is sent-out or generated is unverified.

[F4] The NERC-to-Ember gap is not a stable offset: +8.8% (2022), +11.5% (2023), +1.3% (2024), +6.0% (2025). A stable offset would permit index-splicing the two series; this does not. They are reported side by

side and never merged.
[F5] The connected-population denominator is not computed. Tier 1's correct denominator is connected population, not total — using total folds access expansion into organic growth, the exact conflation Phase 1.6 exists to remove. Requires WDI `EG.ELC.ACCS.ZS`, not yet gathered. Computing it would *lower* the per-capita growth rate further, since the connected share rose over the period, so the headline conclusion is conservative as it stands.
[F6] 2026 Q1 (8,883.47 GWh) is a partial year and is excluded.

To strengthen this

Complete the NERC annual series 2017–2022 from the quarterly reports. This would permit the diagnostic on the authoritative series alone and retire dependence on Ember for shape.
Add WDI `EG.ELC.ACCS.ZS` and recompute on connected population — closes [F5].
Resolve [F2] and [F3] against the source documents.